

Transformers in Practice: BERT, GPT, and Beyond

Scaler School of Technology | Deep Learning I | Week 7 — Session 2/2 | 15 May 2026



What Is This Lecture About?

Last session we built the Transformer architecture from scratch. Today we see how the world actually *uses* it — through two groundbreaking models (BERT and GPT) that changed NLP forever, plus the tokenization systems that make them work.

1 BERT & the Pre-training Revolution

How Google's BERT learns language by filling in blanks (Masked Language Modeling), then adapts to any NLP task with minimal data. *Think of it as medical school (expensive, once) followed by a quick residency (cheap, per specialty).*

2 GPT & Autoregressive Generation

How OpenAI's GPT generates text one token at a time, and why scaling from 117M to 175B parameters unlocked abilities nobody programmed. *Like your phone's autocomplete — but trained on the entire internet.*

3 Tokenization & Practical Considerations

How text becomes numbers (BPE, WordPiece), why LLMs can't do math, and how to choose the right model for your task. *The tokenizer is the most overlooked part — but it determines what the model can even see.*

What to Expect in the Session

- Live demo: BERT predicting masked tokens via HuggingFace
- Live demo: GPT-2 text generation with different temperatures
- ★ See why “15213” breaks LLM arithmetic (tokenizer playground)
- Two quizzes testing BERT vs GPT understanding
- Discussion: “1000 emails to classify — what do you use?”
- Hands-on notebook with BERT fine-tuning for sentiment

How to Prepare (Optional but Helpful)

- ✓ **Recall** the Transformer encoder and decoder from last session — what does each do?
- ✓ **Think about:** what's the difference between understanding a sentence and generating one?
- ✓ **Skim** the HuggingFace Transformers landing page (huggingface.co/transformers) if curious
- ✓ **Brush up on** softmax and attention masks if they felt shaky last time

Duration: ~2 hours **Prerequisites:** Transformer architecture (Week 7, Session 1) **Instructor:** Priyansh Saxena